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Strategic Evaluation of PFAS Disposal Technologies: Unlocking Long-Term Containment and Risk Reduction Through Deep Well Injection

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Abstract

The disposal of per- and polyfluoroalkyl substances (PFAS) continues to challenge both industry and regulators due to the compounds' exceptional persistence and mobility. This study presents a comparative evaluation of three prevailing disposal routes, deep well injection, engineered landfilling, and high-temperature thermal destruction, through the lens of long-term containment performance, environmental risk, and life-cycle cost efficiency. The analysis is intended to inform strategic waste management decisions in anticipation of increasingly stringent regulatory mandates.

The assessment integrates empirical performance data with life-cycle modeling to evaluate each technology's resilience under operational and environmental stressors. Criteria include contaminant isolation efficacy, ease of compliance monitoring, energy and resource demands, and potential downstream liabilities. While landfilling is susceptible to leachate breakthrough and thermal destruction, which entails high operational complexity, deep well injection, when executed in properly characterized and engineered geologic settings, offers passive containment with limited surface exposure and lower long-term oversight burdens.

Results show that deep well injection, supported by rigorous geologic screening and real-time monitoring systems, achieves high containment reliability with comparatively low lifecycle costs. The method's ability to sequester PFAS below potable aquifers reduces exposure pathways and future remediation risks. Furthermore, its low surface footprint and operational simplicity, relative to combustion-based solutions, contribute to its appeal as a scalable long-term strategy.

By framing PFAS disposal through a systems-level perspective, this work positions deep well injection as a viable cornerstone in sustainable hazardous waste management. It offers a practical decision-making framework that aligns environmental safeguards with economic prudence, helping stakeholders navigate technical feasibility, regulatory compliance, and long-term stewardship responsibilities in a complex policy landscape.

Introduction

In recent years, there has been a considerable interest in understanding the environmental distribution, fate, toxicity, and impact of per- and polyfluoroalkyl substances (PFAS), which constitute a large family of fluorinated chemicals, exceeding several thousand that might be in commercial use or spread in the environment, which vary widely in their chemical and physical properties (US EPA, 2024a; US APE, 2024b; CHES, 2023; US EPA, 2020a). The number of PFAS and their uses have expanded over the years. The PFAS family may include more than 12,000 chemical substances that have been used since the 1940s for their favorable characteristics, such as heat, stain, and water resistance, which are desired by industry and consumers across various industrial, commercial, and consumer applications, for example, firefighting foams, semiconductor etchants, non-stick cookware, and stain- and water-repellent textiles. Eventually, they are either directly released into wastewater or surface water through losses from consumer products or generated by the microbial degradation of other per-fluorinated compounds (Phillip, 2025; Gluge et al., 2020; Paul et al., 2009).

Over the recent decades, scientific, regulatory, and public concerns have emerged about potential health and environmental impacts associated with chemical production, product manufacture, and use and disposal of PFAS-containing waste, with evidence increasingly demonstrating that many PFAS compounds exhibit broad-ranging toxicity (Takeda et al., 2023; Hekster et al., 2003). These concerns have led to efforts to reduce the use of or replace certain PFAS, such as certain long-chain perfluoroalkyl carboxylates, long-chain per-fluoroalkane sulfonates, and their precursors, including two widely produced, commonly encountered, and most studied compounds: perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA) (WA DER, 2017; US EPA, 2016).

The primary utility of the PFAS substances stems from the presence of strong carbon-fluorine (C-F) bonds, among the most stable in organic chemistry, which confer exceptional chemical, thermal, and microbial stability. This stability has earned PFAS the moniker "forever chemicals," as they resist degradation and persist in the environment for decades (Suja et al., 2009). The two most studied and detected forms of PFAS are PFOS and PFOA, which tend to have a slow metabolization rate with half-lives of 5.4 years and 3.8 years, respectively, in human blood (Macheka, 2018; Kudo, 2015), with their chemical structures shown in Figures 1 and 2, respectively.

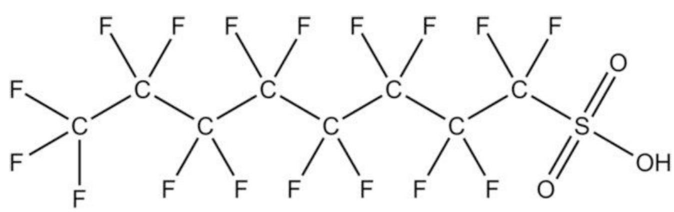

Figure 1—Chemical structure of perfluorooctane sulphonate (PFOS) (Macheka, 2018)

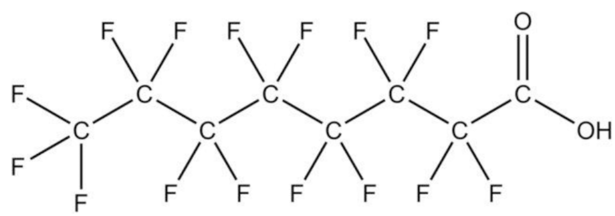

Figure 2—Chemical structure of perfluorooctanoic acid (PFOA) (Macheka, 2018)

Health Risks

Widespread use and improper disposal of such PFAS materials and PFAS-contaminated materials have resulted in their detection in air, soil, water, in addition to the aquatic ecosystems, and not only in specific industries but also worldwide, stringently increasing the need for regulatory thresholds and disposal policies for PFAS containment worldwide. According to a report by the Centers for Disease Control and Prevention (2024), most people in the US have been exposed to PFAS, and nearly all people in the United States (US) have it in their blood, highlighting the pervasive nature of contamination and resilience (CDCP, 2024). Public health concerns and regulatory scrutiny increased due to environmental persistence, bio-accumulative potential, and associated health risks, such as endocrine disruption (Lewis et al., 2024; Arora et al., 2023), carcinogenicity (Zheng et al., 2024; Singh and Hsieh, 2021), developmental effects, and immune effects (Liew et al., 2018; Pennings et al., 2016; Stein et al., 2016).

Worldwide Concerns

By way of example, in Korea, PFOS concentrations ranging from 2.24 to 651 ng/L and PFOA concentrations ranging from 0.9 to 62 ng/L have been reported in Lake Shihwa, which were among the highest ever measured in the environment (Yoo et al., 2009; Yoo et al., 2008; Rostkowski et al., 2006). In Sweden, the guideline value of drinking water is set at 90 ng/L, reflecting a precautionary approach to human health protection (Eriksson, 2025; Gobelius et al., 2018; Worley et al., 2017). In the US, specifically in Maine, the state has taken significant steps to better protect, conserve, and enhance its natural resources and environmental quality, where by April 2022, the Maine Department of Environment Protection (MDEP) Legislature passed a law prohibiting the land application of biosolids (sludge and sludge-derived compost) containing PFAS, as well as their sale as compost or agricultural materials making Maine the first state in the US. to prohibit spreading PFAS-laden sludge as fertilizer (MDEP, 2022).

History of the EPA Regulations

At the US federal level, over the past decade, the US Environmental Protection Agency (US EPA) has progressively strengthened its regulatory framework to address the growing concerns associated with PFAS contamination. The earliest significant step was the proposal to designate particular high-fluorine "chemical" wastes as hazardous under the comprehensive Resource Conservation and Recovery Act (RCRA), Subtitle C, marking the beginning of a concerted regulatory effort (Pollack et al., 2024; US EPA, 2015). This initiative laid the groundwork for subsequent developments in PFAS management. In 2019, the US EPA issued a notice to solicit public and industry input on including PFAS chemicals in the Toxic Release Inventory (TRI), thereby increasing transparency regarding their environmental releases (US EPA, 2019a; US EPA, 2019b). The following year, in 2020, the agency finalized regulations to add 172 PFAS chemicals to the TRI list. It imposed restrictions on their import and sale without prior US EPA approval, signaling a shift toward stricter control and reporting requirements (US EPA, 2020b). Building on these efforts, 2021 saw the launch of a comprehensive PFAS Testing Strategy, which included nationwide monitoring efforts, particularly in drinking water sources, and the proposal of a Toxic Substances Control Act (TSCA) Section data-gathering rule designed to improve the collection of detailed information on PFAS usage, release, and environmental presence (US EPA, 2021a).

In 2022, the EPA issued health advisories for specific PFAS compounds, providing critical guidance to protect public health (Gaballah et al., 2020; US EPA, 2022; Tian and Sun, 2019). The agency's enforcement efforts intensified in 2023 through National Enforcement and Compliance Initiatives, although actions against certain companies, such as Inhance Technologies LLC, faced legal challenges and were later vacated by courts (Anaya et al., 2023; US EPA, 2023; Dervisevic, 2022). By 2024, the EPA introduced recommended water quality criteria for aquatic life, proposed effluent discharge guidelines for PFAS manufacturing facilities, and included PFAS on the Monitoring List for fish and shellfish advisories (Giffard et al., 2022). The PFAS Designation Rule also became effective in July 2024, establishing legal standards for PFAS regulation, despite ongoing litigation (US EPA, 2024b; US EPA, 2024c). This regulation would redirect many waste streams currently disposed of in municipal landfills or Class II saltwater wells into more costly, specialized hazardous waste treatment facilities (US EPA, 2025). Also, the US EPA's 2023 and 2024 Interim Guidance explicitly favors underground injection control (UIC) Class I injection wells as a "long-term, passive" disposal method for high-concentration PFAS liquids. (US EPA, 2025; US EPA, 2024b; US EPA, 2023; US EPA, 2021b). This signals a policy shift towards injection technology for permanent containment, with future regulations expected to expand the use of well injection as a preferred strategy for PFAS risk reduction and long-term environmental safety (Hagarty, 2023).

In April 2024, the EPA finalized enforceable Maximum Contaminant Levels (MCLs) for six PFAS compounds in drinking water, a landmark federal regulatory action. The MCL for both PFOA and PFOS became 4 ng/L, levels that are orders of magnitude lower than the previous health advisory level of 70 ng/L, emphasizing the increasing regulatory focus on PFAS contamination and long-term public health safety (Paustenbach et al., 2025; US EPA, 2024b; Webster, 2024). Most recently, in 2025, the EPA announced it would rescind or reconsider MCLs for four of those compounds, retaining only PFOA and PFOS standards and extending compliance timelines, marking a significant milestone in federal efforts to reduce human exposure, protect environmental resources, and establish a robust regulatory framework to manage this complex class of persistent chemicals (Paustenbach et al., 2025; Hua et al., 2025).

Objective of This Paper

The increasing stringency of regulatory standards for PFAS disposal underscores the urgent need for effective and sustainable management strategies that ensure long-term containment while minimizing risks to human health and the environment. Determining the appropriate method for the ultimate disposal of PFAS is a complex issue due to their volatility, solubility, environmental mobility, and persistence. Waste producers and regulators must choose from three leading, well-established, commercially available disposal options: high-temperature thermal treatment, engineered landfilling, and deep injection disposal wells. Additionally, several emerging niche treatments are being considered as potential alternatives.

This paper provides a comprehensive overview of these disposal techniques, analyzing and comparing them based on multiple decision-making criteria, including cost, energy efficiency, environmental impact, and long-term containment efficacy. Building on this, the subsequent sections demonstrate why properly engineered deep-well injection stands out as the most economical, energy-efficient, and environmentally protective solution for the large-scale management of PFAS-contaminated materials.

Methods Available for PFAS Disposal

Thermal Treatment

Thermal treatment technologies have emerged as leading approaches for breaking down PFAS compounds, aiming to convert them into less harmful end products such as carbon dioxide, water, and hydrogen fluoride. One of the primary methods is high-temperature incineration. Other methods are being developed, such as plasma treatment, supercritical water oxidation, and thermal desorption. Secondary treatment offers pathways to destroy PFAS in different waste streams, though each requires carefully controlled conditions to prevent the formation of toxic byproducts.

High-temperature incineration remains the best-documented route for outright PFAS destruction. It aims to break PFAS down by combusting them at very high temperatures, typically exceeding 1,000 °C, to break the strong carbon-fluorine bonds. For example, unimolecular thermal destruction modeling indicates that carbon tetrafluoride (CF_4), one of the most robust fluorinated compounds, requires temperatures above 1,400 °C to achieve 99.99 % destruction (Challa Sasi, 2022). Limited studies on the ability of thermal destruction of fluorotelomer-based polymers found no detectable levels of PFOA after a two-second residence time and about 1,000 °C (Longendyke et al., 2022) In practical applications, a recent study at a hazardous-waste facility in Port Arthur, Texas, reported PFAS destruction efficiencies of over 99%, with PFOS and PFAS reaching up to 99.9999%, by reaching a secondary combustion chamber temperature of 2,040 °F and a residence time of 2.3 seconds, with minor solid and liquid residue amounts below the US EPA drinking water limit. Such studies suggest a high potential for complete breakdown; however, they depend critically on maintaining optimal conditions like sufficient oxygen, temperature, and residence time.

Despite these high efficiencies, several significant drawbacks undermine confidence in the method. First, incomplete combustion can lead to unknown and potentially smaller hazardous PFAS byproducts, or products of incomplete combustion (PICs), in addition to smaller PFAS fragments that may be equally persistent (Krug et al., 2022). Secondly, there is a high risk of significant carbon emissions. Incinerating PFAS-laden aqueous film-forming foam (AFFF) can be highly carbon-intensive, where the complete combustion may emit approximately two metric tons of CO? per ton of AFFF. In comparison, improper combustion scenarios could spike emissions to yield the equivalent impact of 439-537 metric tons of CO? per ton Yao et al., 2022; Wang et al., 2021). Third, the high disposal cost, based on information summarized in recent US EPA guidance, is that disposal costs via high-temperature incineration are around $1,000 to $2,000 per ton of material, making incineration the most expensive disposal method among available options (US EPA, 2020c). Finally, regulatory and measurement challenges remain. Existing studies often suffer from high detection limits or test only tiny PFAS quantities, so the real-world effectiveness and byproduct characterization remain poorly understood.

Engineered Landfilling

Engineered landfilling manages PFAS primarily by containment, not destruction. Modern regulated municipal solid waste landfills (MSWL) use composite liners, leachate collection, and covers to limit any breach and capture a concentrated leachate stream for treatment. When PFAS-containing waste (such as consumer products, packaging, or biosolids) is deposited in lined landfills, the stream is divided into leachate, landfill gas condensate, and occasionally surface runoff. PFAS are highly soluble and mobile in water; thus, leachate acts as the dominant transport medium. Lang et al. (2017) estimated that leachate generation in the US is 61.1 billion liters annually (Lang et al., 2023). In some US studies, PFAS content in MSWL leachate ranged below detection levels and up to 125,000 ng/L, with an average of 10,500 ng/L. According to studies by Tolaymat 2023, it is estimated that 84% of PFAS feed to MSWL remains in waste mass, while 5% leaves via LFG and 11% via leachate annually (Tolaymat et al., 2023).

Another recent critical review data report PFAS concentrations in landfill leachate up to 48,700 ng/L. Although landfill leachate typically represents only ~1.7% of influent flow to wastewater treatment plants (WWTPs), it accounts for approximately 18% of the total PFAS mass load (Masoner et al., 2020). This containment-and-concentration principle is the chief functional advantage of engineered landfills: they capture PFAS from heterogeneous waste into relatively controlled streams that can be monitored and treated.

The disadvantages are substantial. First, landfilling does not destroy PFAS, while leachate and off-gas side-streams can mobilize them. Gas-system condensate carried PFAS at a median of 7,000 ng/L (Chen et al., 2024; Chen et al., 2023; Feng, 2020), comparable to leachate, indicating volatilization/partitioning pathways within the waste mass. Second, long-term containment is uncertain because PFAS can slowly pass through landfill liners, even those made of polyethylene. Studies show that tiny flaws or pinholes in HDPE liners can let PFAS leak much more easily, raising the risk of long-term escape (Moore et al., 2024; Battista et al., 2020). Even with double-composite liners, transport modeling still evaluates breakthrough risk over time, not elimination (Barakat et al., 2025; Gates et al., 2020). Finally, on-site treatment can remove most of PFAS from leachate but generates PFAS-rich concentrates that require further management, and breakthrough in leachate matrices is faster than in drinking water, which raises cost and liability without resolving ultimate disposal.

Reflecting these concerns, the US EPA has begun rulemaking to establish PFAS effluent and pretreatment standards for landfills discharging leachate, acknowledging that containment strategies alone leave open long-term release pathways and liability risks. In addition, one proposed rule would add nine PFAS compounds to the list of hazardous constituents under RCRA. At the same time, another would broaden EPA's Corrective Action authority to cover emerging contaminants such as PFAS explicitly. If these efforts ultimately lead to the classification of certain PFAS-rich liquids as a Subtitle C hazardous waste, landfill operators could face significantly higher disposal costs, and crucially, the potential for long-term liability from liner failures would persist despite stricter regulation (US EPA, 2024a; US EPA, 2024b; US EPA, 2024c).

Deep-Injection Disposal Wells

Deep injection disposal wells are among the few PFAS disposal options for isolating large volumes of PFAS-contaminated waste far deeper than any drinking-water aquifers. The US EPA's 2024 update to its Interim Guidance explicitly lists underground injection alongside landfilling and thermal treatment as current technologies used for disposal and notes that underground injection wells "are designed to isolate liquid wastes deep below the land surface and ensure protection of underground sources of drinking water" with a lower potential for environmental release than other alternatives. Deep well injection is the only option to maintain a permanent solution by containment when it's cost-effective and available (US EPA, 2025; Feldman, 2024; US EPA, 2021b).

From an engineering and regulatory standpoint, underground injection hazardous wells operate under quantifiable safeguards in the United States. Before injection, operators must show that hazardous materials will not migrate from the injection zone for 10,000 years, and hazardous wells use a 2-mile area-of-review (AOR) versus a quarter mile for non-hazardous wells (US EPA, 20012001). During operations, federal rules require periodic mechanical-integrity testing: annual pressure tests of the tubing/casing system, an annual radioactive tracer survey to verify cement at the injection interval, and at least one temperature or noise log every five years to check for vertical fluid movement; continuous surface monitoring of injection pressure, rate, and volume is also required (US EPA 1980). Injection pressures are kept below the rock's fracture gradient, which field data from sites spread through Michigan, Ohio, and Indiana show a fracture gradient range of 0.48 to 0.91 psi/ft in US Class I wells, another quantified margin against upward migration (US EPA 1994).

Underground Injection Control (UIC) Class I hazardous-waste wells sequester liquid PFAS waste at depths ranging from 1,700 ft to 10,000 ft or deeper in permeable injection zones overlain by an impermeable cap-rock as confining strata, as shown in Figure 3. The confining layer may be associated with additional impermeable layers of rock, salt beds, and dense shales to separate the injection zone from the drinking water zones (US EPA, 2025; US EPA, 2021b). Three to five concentric steel casings are cemented in place, and continuous annulus-pressure monitoring provides real-time integrity verification. In one of the major commercial sites, the operator reported injecting more than 19 million barrels of waste volumes in a Class II deep injection well in Kenedy, Texas. Operators at another commercial site in Deer Park, Texas, have managed over 1.25 billion pounds of PFAS-contaminated waste since 2017 with no surface or groundwater discharge (Magee, 2025). Historically, the US EPA concluded that Class I hazardous injection wells are "safer than virtually all other waste disposal practices," a finding based on risk comparisons across regulated disposal methods. These quantitative depth, volume, and risk benchmarks help explain why deep wells are increasingly selected for PFAS brines and leachate concentrates.

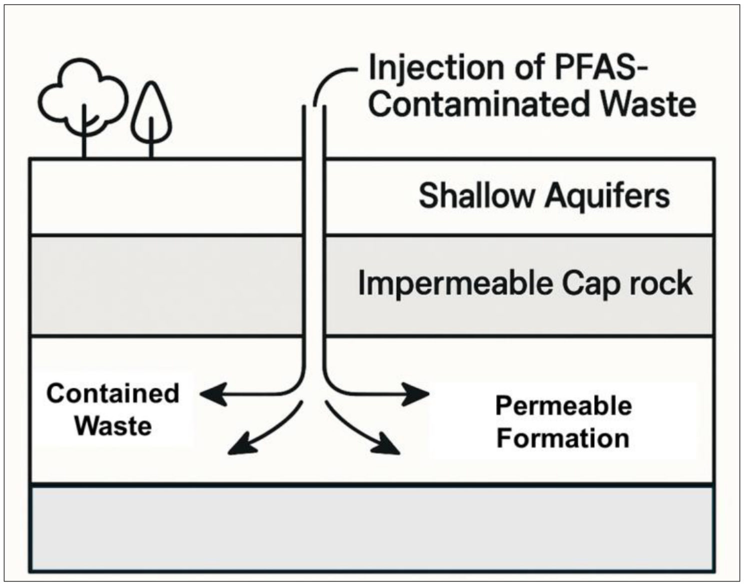

Figure 3—Injection of PFAS-contaminated waste in a deep well within a permeable formation capped with an impermeable formation

The deep well injection does not destroy PFAS; it relies on long-term geologic confinement and continued regulatory oversight, and it is not available everywhere with suitable geology (US EPA notes availability constraints). Modeling in the US EPA's risk study shows that failure scenarios (e.g., an abandoned borehole short-circuit) can elevate risk depending on permeability contrasts, underscoring the value of the 2-mile AOR. Preparing a robust No-Migration Petition typically involves substantial technical work (US EPA reports ~11,000 staff hours and costs exceeding $2 million, though simpler cases average ~$300k), and wells must maintain integrity under recurring tests. Even with these caveats, when a permitted Class I facility is available and the waste is an aqueous PFAS stream, the measured depth of disposal, quantified pressure margins, 10,000-year modeling standard, and continuous MIT and monitoring regime make deep well injection the most controllable end-of-pipe option currently deployable at scale, preventing the contamination from spreading to groundwater or surface water sources, preventing surface liabilities, and maintaining environmental and community health in the surrounding area.

Other Remediation Methods

A suite of emerging or supplemental techniques is available for site-specific situations. Bioremediation using anaerobic or co-metabolic environments can partially de-fluorinate carbon-based PFAS, but reaction kinetics remain slow, and field demonstrations are scarce (Bhattacharya., etal. 2025). Stabilization with organophilic clays such as FLUORO-SORB® can immobilize more than 95 % of leachable PFAS but still provides containment rather than destruction and thus carries long-term monitoring obligations (El Mohtar., et al. 2023). Electro-kinetic remediation, an extraction technique based on the principle that, under the effect of a direct current and low-intensity electric field, charged contaminants will transport towards the electrode of opposite charge (Acar et al., 1996; Acar et al., 1995), has removed up to 70% of PFOS and PFOA from fine-grained soils in laboratory columns. Still, efficiency declines in high-organic matrices and electrode effluents require further treatment (Georgios N.,etal. 2022). Phytoremediation or Phytoextraction is the translocation of contaminants in the soil or other media by plant roots into the aboveground portions of the plants. It can reduce pore-water PFAS by 40% over multi-season campaigns, but biomass harvesting and slow kinetics limit scalability (BBiotech, 2024; Surriya et al., 2015). Soil washing and foam fractionation removed 88-96 % of PFAS waste from a 1,000-ton sandy-soil pilot, but it generated a concentrated liquid residual requiring off-site disposal (Uwayezu et al. 2024). While valuable niche tools, none offer the capacity, applicability, or regulatory certainty of the previously mentioned primary routes.

Technical Comparison of PFAS Disposal Methods

Environmental Release Potential

Thermal incineration of PFAS-containing wastes remains controversial, as incomplete combustion can release fluorinated byproducts, including hydrogen fluoride and ultrafine particulates, which pose direct risks to air quality and climate (US EPA, 2024a; US EPA, 2024c). Landfilling likewise generates PFAS-rich leachate and gas condensate, which are prone to liner leakage and long-term migration, raising concerns for groundwater and surface water contamination (LaRoe et al., 2023; Berg et al., 2022). By contrast, deep-well injection isolates liquid wastes thousands of feet below the impermeable confining layers, minimizing risk of drinking water contamination, and the US EPA has identified it as the method with the lowest uncertainty of environmental release (Meegoda et al., 2022; Horst et al., 2020; US EPA, 2016).

Technical Reliability and Uncertainty

Incineration requires very high temperatures (more than 1,100 °C) and advanced air pollution controls, but several studies show incomplete PFAS mineralization under operational conditions, making reliability questionable (US EPA, 2024a; US EPA, 2024c). Landfills, while simple in design, cannot guarantee long-term containment, as liners degrade and leachate systems eventually fail, leaving future generations at risk (LaRoe et al., 2023; Berg et al., 2022). Deep-well injection, by comparison, benefits from decades of regulatory experience under the Underground Injection Control (UIC) program and is regarded by regulators as technically reliable for high-strength PFAS liquids, with low migration potential when wells are constructed and appropriately monitored.

Regulatory Alignment and Acceptance

The regulatory landscape is shifting away from incineration and landfilling. States such as New York and Illinois have enacted restrictions on PFAS incineration, and Maryland has banned both incineration and landfilling of PFAS- laden firefighting foams (SCS Engineers, 2023). Meanwhile, EPA's 2024 Interim Guidance explicitly highlights deep-well injection as a compliant and preferred option within the existing UIC framework, offering regulatory certainty that the other methods lack (US EPA, 2024a; US EPA, 2024c). This regulatory endorsement underscores the growing acceptance of the deep well injection over surface-based disposal.

Capacity and Scalability

Rotary kilns and incinerators typically operate at modest capacity (a few tons per hour) and require periodic shutdowns for refractory maintenance, limiting scalability (US EPA, 2024a; US EPA, 2024c). Landfills can accept large quantities of solids but impose restrictions on liquids to prevent hydraulic overload, severely limiting their role in PFAS leachate and foam disposal (LaRoe et al., 2023; Berg et al., 2022). By comparison, in East Texas, a single Class I deep injection well can handle 5,000 to 15,000 bbl/day, and multi-well manifolds can exceed 50,000 bbl/day, making deep well injection uniquely suited to high-volume liquid PFAS wastes.

Long-Term Containment and Liability

Thermal facilities eventually require decommissioning and pose ongoing liability for emissions during their operational life. Landfills demand federally mandated 30-year post-closure care, but PFAS mobility exceeds that horizon, creating intergenerational liability risks (Uszkiewicz, 2025). However, deep-well injection provides containment within geologic formations for decades, with wells subject to continuous monitoring and testing annually and every five years, after which they are permanently sealed, effectively transferring liability away from the surface environment Meegoda et al., 2022; Horst et al., 2020). This makes deep well injection the most secure and least burdensome option over time.

Community and Environmental Justice Impacts

Incinerators emit pollutants into surrounding communities, and landfills often place greater burdens on nearby residents who face increased exposure to contaminated leachate, odors, and long-term environmental risks. Both methods, therefore, raise long-term concerns regarding environmental justice (LaRoe et al., 2023; Berg et al., 2022). Deep-well injection, by contrast, occurs entirely underground, reducing community exposure to PFAS or combustion byproducts. The US EPA explicitly notes that injection minimizes surface-level impacts and aligns with broader goals of reducing disproportionate pollution burdens. Thus, it represents the most socially responsible option for large-scale PFAS management.

PFAS Challenges in Wastewater Treatment Plants

WWTPs face unique challenges in managing PFAS because the conventional treatment processes, such as activated sludge, aeration, and digestion, are ineffective at breaking down these highly stable fluorinated compounds (US EPA, 2024a; US EPA, 2024c). Instead of being destroyed, PFAS partition into different waste streams: short-chain compounds tend to remain in the plant and pass through to rivers or lakes, while long-chain PFAS build up in biosolids. As a result, WWTPs inadvertently serve as pathways for PFAS redistribution, contributing both to surface water contamination and to land-applied biosolids that can impact soils and groundwater (Arvaniti et al., 2014). Figure 4 shows the byproducts from WWTPs that contain PFAS which can enter surface waters and sediments via treated effluent, migrate to groundwater through infiltration, and reach agricultural soils, and subsequently crops and livestock, when biosolids are land-applied. In addition, volatilization and aerosolization can release PFAS to the atmosphere. Collectively, these routes concentrate exposure pathways that ultimately lead to human intake through drinking water and the food chain. This persistence creates a regulatory and operational dilemma, since effluent and biosolids both remain within existing discharge and reuse cycles.

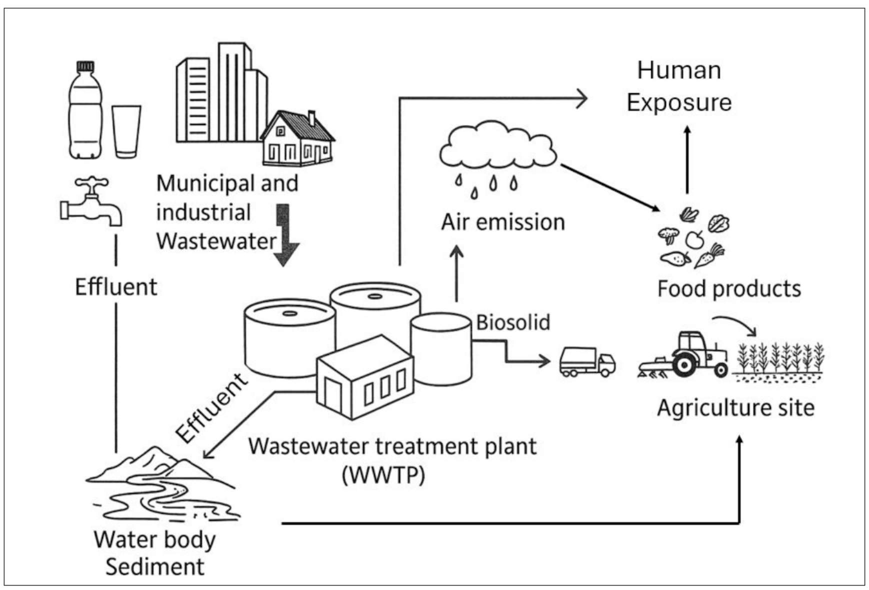

Figure 4—Environmental and Human-Exposure Pathways of WWTP-Derived PFAS

To address these challenges, WWTPs have increasingly turned to advanced treatment and disposal methods. Technologies such as granular activated carbon (GAC), ion exchange resins, and high-pressure membrane processes (nanofiltration and reverse osmosis) can effectively capture PFAS from wastewater streams. However, they produce PFAS-concentrated brines requiring final disposal (Cao et al., 2025). Destructive technologies, including supercritical water oxidation, electrochemical oxidation, and plasma treatment, are under development to mineralize PFAS at the concentrated stage, though they are not yet widely deployed at scale. In parallel, deep-well injection has gained traction as a near-term mitigation strategy for liquid residuals, offering a secure and regulated endpoint while more permanent destruction technologies continue to mature (US EPA, 2024a; US EPA, 2024c).

Case Study - Terminal Island Renewable Energy Project (TIRE)

To address this challenge, a case study is presented that highlights the role of deep-well injection as a secure and effective disposal method for PFAS-contaminated wastewater and biosolid-derived residuals from wastewater treatment plants, specifically, the Terminal Island Renewable Energy (TIRE) project at the Los Angeles Terminal Island Water Reclamation Plant (TIWRP). This is the first full-scale US facility that permanently injects municipal biosolids, treated sludge, and reverse-osmosis brine into deep geologic sandstone formations at a depth of 5,000 to 7,500 ft, allowing natural heat and pressure to drive anaerobic biodegradation into methane and CO?, thereby securely and permanently sequestering both organic matter and greenhouse gases (US City of Los Angeles, 2011). Since July 2008, TIRE has now handled 100% of Terminal Island Plant's biosolids and roughly 20% of Hyperion Treatment Plant's residuals, and it has injected about 1 billion gallons of biosolids slurry. (GeoEnvironment-Technologies, 2025).

TIRE's operational safety is ensured through rigorous monitoring, including down-hole pressure sensors, fiber-optic temperature measurement, periodic geophysical logging, and sampling from offset monitoring wells to confirm containment within the target formation. By eliminating surface disposal, trucking, and land application, the project achieves substantial environmental gains: enhanced protection of surface and groundwater, reduced energy use, and lower greenhouse gas emissions. Crucially, the facility injects reverse-osmosis brine, which commonly contains PFAS, effectively isolating PFAS in a secure, subsurface reservoir. The geological formation, well design, and regulatory framework at TIRE provide a scalable and verifiable model for safely disposing of PFAS-laden liquids, sludges, and biosolids in line with the US EPA's interim PFAS disposal guidance, while delivering high capacity and a minimal surface footprint.

Summary and Conclusion

This paper evaluates three commercially available PFAS disposal routes: high-temperature thermal treatment, engineered landfilling, and deep-well injection, against environmental release potential, technical reliability, capacity, life-cycle cost, regulatory fit, and community impacts. PFAS persistence and mobility make conventional wastewater and solids management insufficient, and regulatory tightening (e.g., federal interim guidance and drinking-water limits) elevates the importance of secure, auditable endpoints. Across criteria, deep-well injection performed best when implemented in properly selected formations under the US EPA UIC program: it isolates high-strength PFAS liquids far below potable aquifers, operates with real-time integrity monitoring and periodic mechanical integrity tests, and has the lowest uncertainty of environmental release among current options (US EPA 2024c). In contrast, thermal treatment can achieve high measured destruction efficiencies only under narrow operating windows while still facing emissions/byproduct uncertainty and high cost, and engineered landfilling concentrates PFAS into leachate and gas condensate streams that preserve long-term release pathways and liabilities.

The analysis integrates system-level considerations at WWTPs, where PFAS commonly pass through conventional processes and partition into effluent and biosolids. Advanced capture can remove PFAS but generates concentrated residual liquids that require a secure endpoint; deep-well injection provides that endpoint today while destructive technologies continue to mature. The TIRE case demonstrates long-duration, full-scale feasibility: since 2008, four deep wells at 5,000-7,500 ft have managed TIWRP biosolids and sludge with continuous monitoring and documented confinement, providing high throughput with a small surface footprint and aligning with EPA interim guidance. Together, the comparative assessment and the TIRE evidence indicate that deep-well injection offers durable containment, lower life-cycle cost and oversight burden, and a practical route to reduce exposure pathways and future remediation risk.

Deep-well injection is the most controllable and scalable "end-of-pipe" option currently deployable for PFAS-laden liquids, leachates, and brines, provided that (i) geologic evaluation confirms a permeable injection zone under strong confining strata, (ii) UIC engineering and monitoring standards are met, and (iii) operations remain within allowable pressure limits and mechanical integrity is verified on the required basis. Under these conditions, deep well injection reduces surface contact and environmental release uncertainty more effectively than combustion-based or surface-containment pathways, while offering higher sustained throughput and lower life-cycle cost and liability. Engineered landfilling should be reserved for solids only and paired with stringent leachate controls, recognizing long-term breakthrough risk. Thermal destruction may be suitable for select waste streams where validated operating conditions, emissions characterization, and cost allow. However, it should not be considered a universal solution given operational complexity and byproduct concerns.

For WWTPs and industrial generators, a practical strategy is to (1) capture PFAS with proven separation processes, (2) route the concentrated residuals to deep well injection where available, and (3) continue piloting destructive technologies on the concentrated side-streams as they mature. Policy and procurement should prioritize permitting of regional Class I capacity in suitable basins; standardize no-migration demonstrations and area-of-review practices; and align pretreatment and effluent rules to direct high-strength PFAS liquids away from municipal landfills and toward secure subsurface isolation. Future work should refine long-term fate modeling, expand tracer-based confinement verification, and develop clear decision points for when to pivot from containment to destruction as technologies reach commercial readiness. In sum, deep-well injection, implemented to UIC standards, provides the strongest near-term foundation for PFAS risk reduction and long-term containment, while preserving flexibility to integrate emerging destructive methods as they become proven at scale.

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